Artificial Intelligence

Artificial intelligence, or AI, is the broad goal of building computer systems that perform tasks that normally need human intelligence, such as understanding language, recognising images and making decisions. Machine learning is a subset of AI, and deep learning is a subset of machine learning.

Neural Networks and Deep Learning

Deep learning uses neural networks with many layers. A neural network is made of neurons that take weighted inputs, add a bias and apply an activation function. Training adjusts the weights using backpropagation, which computes gradients of the loss with respect to every weight.

Deep learning is especially strong on images, audio and text, where hand-designing features is difficult.

Transformers, LLMs and RAG

The Transformer is a neural network architecture built around self-attention. Self-attention lets every word in a sentence look at every other word and decide which ones are important for its meaning. Transformers were introduced in 2017 and are the foundation of modern large language models.

A large language model, or LLM, is a Transformer trained on huge amounts of text to predict the next token. LLMs can hallucinate, which means they sometimes state false information confidently.

Retrieval-Augmented Generation, or RAG, reduces hallucination by first retrieving relevant passages from a trusted document collection and then asking the LLM to answer using only those passages. RAG also lets a system use private or recent documents without retraining the model.